Chapter #7 (C.F Mgt)

Date

(7-16): Equilibrium Stock Price:-

$R_{rf} = 11\% = 0.11$ (Risk free rate)

$R_m = 14\% = 0.14$ (Market required return)

$\beta = 1.5$

a) what price should scheular sell stock:-

if $D_1 = 2.25\$$, $g = 0.05$ what is P_0

$\therefore P_0 = \frac{D_1}{r - g}$ → where $r = R_{rf} + \beta(R_m - R_{rf})$ ↓ CAPM

$$P_0 = \frac{2.25}{(0.11 + 1.5(0.14 - 0.11)) - 0.05}$$

$$\boxed{P_0 = 21.43\$}$$

b) FRB increases money supply now

$R_{rf} = 9\%$ and $R_m = 12\%$

$$P_0 = \frac{2.25}{(0.09 + 1.5(0.12 - 0.09)) - 0.05}$$

$$\boxed{P_0 = 26.47\$}$$

Date:

c) Investor's risk aversion declines,

$R_m = 11\% = 0.11$

Now $P_0 = ?$

$$P_0 = \frac{2.25}{(0.09 + 1.5(0.11 - 0.09)) - 0.05}$$

$$\boxed{P_0 = 32.14\ \$}$$

d) Change in management.

$g = 6\%$, $\beta = 1.3$, $R_{rf} = 0.09$, $R_m = 0.11$

$D_1 = 2.287$, $P_0 = ?$

$$P_0 = \frac{2.27}{(0.09 + 1.3(0.11 - 0.09)) - 0.06}$$

$$\boxed{P_0 = 40.54\ \$}$$

Date:

(7-17):- Constant growth Stock valuation:-

common stock paid dividend $D_0 = \$2$ yesterday
$g = 5\%$ for next 3 years
Plan to buy the stock and hold it for 3 years and then sell it.

a) expected dividends for next 3 years
$D_1, D_2, D_3 = ?$

$D_1 = D_0(1+g) \Rightarrow 2(1+0.05) \Rightarrow \boxed{D_1 = 2.1}$
$D_2 = D_1(1+g) \Rightarrow 2.1(1.05) \Rightarrow \boxed{D_2 = 2.205}$
$D_3 = D_2(1+g) \Rightarrow 2.205(1.05) \Rightarrow \boxed{D_3 = 2.31525}$

b) Given discount rate $r = 12\%$.
ΣPV of $D_1, D_2, D_3 = ?$

$$PV(D_t) = \frac{D_t}{(1+r)^t}$$

$$\Sigma PV = \frac{2.1}{(1+0.12)^1} + \frac{2.205}{(1+0.12)^2} + \frac{2.31525}{(1+0.12)^3}$$

$$\Sigma PV = 1.875 + 1.757 + 1.6479$$

$$\boxed{\Sigma PV = 5.280}$$

Date:

c) expected stock price to 3 years from now P_3 34.73 . $\Rightarrow P_3 = 34.73\$$

$r = 12\%$ discounted rate

PV of P_3 = ?

$$\therefore PV(P_t) = \frac{P_t}{(1+r)^t}$$

$$PV(P_3) = \frac{34.73}{(1+0.12)^3}$$

$$\boxed{PV(P_3) = 24.720\$}$$

d) if you plan to sell the stock after 3 years of 34.73$, what should you most pay for it today:-

$$\text{Max Value} = \underbrace{PV(D_1) + PV(D_2) + PV(D_3)} + PV(P_3)$$

$$= 5.280 + 24.720$$

$$\boxed{\text{Max Val.} = 30.\$}$$

Date:

e) Use Equation 7-2 (Gordon) to compute present value (P_0) where $g = 5\%$ constant.

$$\therefore P_0 = \frac{D_1}{r_e - g} \Rightarrow P_0 = \frac{2.1}{0.12 - 0.05}$$

$P_0 = 30\ \$$

f) Is the value of this stock dependent on how long to hold it. i.e. if you plan to hold it for 2 years of 5 years, what would be the P_0 (today's stock value).

Ans:- In constant growth present value would not be affected because its the sum of all PVs of dividends including the terminal sale price if you sell in any future time.

Date:

(7-18) Non-constant growth S.V:-

Most recently paid dividend yesterday $D_0 = 1.75$

RT for next 5 years $\Rightarrow g_{1-5} = 15\%$

after 5 years growth slows down to 5% per year $\Rightarrow g = 5\%$

$r = 12\%$ (required rate)

a) RT's expected dividends for $t=1$, $t=2$, $t=3$, $t=4$, $t=5$

$D_1 = D_0(1+g) \Rightarrow D_1 = 1.75(1+0.15) \Rightarrow D_1 = 2.0125$

$D_2 = D_1(1+g) \Rightarrow D_2 = 2.0125(1.15) \Rightarrow D_2 = 2.3144$

$D_3 = 2.3144 \times 1.15 = 2.6615$

$D_4 = 2.6615 \times 1.15 = 3.0608$

$D_5 = 3.0608 \times 1.15 = 3.5199$

b) intrinsic value of the stock today (P_0) = ?

PVs of $P_1, P_2, P_3, P_4, P_5, P_5$

$$\sum PV = PV(P_1) + PV(P_2) + PV(P_3) + PV(P_4) + PV(P_5) + PV(P_5)$$

$$\therefore P_5 = \frac{D_6}{r-g} \quad \therefore PV(P_t) = \frac{P_t}{(1+r)^t} \quad \therefore PV(P_t) = \frac{P_t}{(1+r)^t}$$

$$\sum PV = \frac{2.0125}{(1+0.12)^1} + \frac{2.3144}{(1.012)^2} + \frac{2.6615}{(1.12)^3} + \frac{3.0608}{(1.12)^4} + \frac{3.5199}{(1.12)^5} + \frac{\frac{3.5199(1.05)}{0.12-0.05}}{(1.12)^5}$$

Date:

~~$P_1 = 36.23$ $~~

$= 9.47878 + 29.95928684$

so,

$P_0 = 39.438$ $

c) expected dividend yield = ?
capital gain yield in 1st year = ?
expected total return = ? (required rate)
also calculate these three for t=5

* for year 1:-

∴ dividend yield = D_1/P_0

$= 2.0125/39.438$

dividend yield = 0.05105 or 5.105%

∴ capital gain yield = $\frac{P_1 - P_0}{P_0}$

$= \frac{\left(\frac{D_1}{r-g}\right) - 39.438}{39.438}$

$= \frac{\left(\frac{2.0125}{0.12-0.05}\right) - 39.438}{39.438}$

∴ Total return = dividend yield + capital gain yield

0.12 = 0.05105 + capital gain yield

capital gain yield = 0.12 − 0.05105

Date:

Capital gain yield = 6.895 %

* for year 5 :-

dividend yield : D_6/P_5

$$\text{dividend yield} = \frac{(3.5199 \times 1.15)}{[3.5199 \times 1.15 / (0.12 - 0.05)]}$$

dividend yield = 0.07 or

dividend yield = 7%

Capital gain yield = total return − dividend yield

CGY = 12% − 7%

Capital gain yield = 5%

so, at t=1

r = 12%, CGY = 6.895%, DY = 5.105%

at t=5

r = 12%, CGY = 5%, DY = 7%

Date:

d) If your intrinsic value differs substantially from market price what would happen?

Investors would buy or sell until equilibrium has established by reaching of market price to intrinsic value.

(7-10) Supernormal growth S.V:-

$g = 20\%$ (for recent 2 years)
supernormal growth for another 2 years:-

$g_1 = 20\%$ $g_2 = 20\%$

a) Expected dividend yield = ?
Capital gain yield = ?

$D_0 = 1.60\$$, $g_L = 6\%$

$r_s = 10\%$

$P_0 = ?$ P.V = ?

for 2 years

$P.V = P.V(D_1) + P.V(D_2) + P.V(P_2)$

$D_1 = 1.60(1+0.20) = 1.92$

$D_2 = 1.92(1+0.20) = 2.304$

$D_3 = 2.304(1.06) = 2.4422$

$P.V = \frac{1.92}{(1+0.10)^1} + \frac{2.304}{(1+0.10)^2} + \frac{(1.4056256/0.10-0.06)}{(1+0.10)^2}$

P.V = 58.8207 $

for 2 years :-

$\therefore P.V = PV(D_1) + PV(D_2) + PV(P_2)$

$D_1 = 1.60(1+0.20) = 1.92$

$D_2 = 1.92(1+0.20) = 2.304$

$D_3 = 2.304(1+0.06) = 2.4442$

$$PV = \frac{1.92}{(1+0.10)^1} + \frac{2.304}{(1.010)^2} + \frac{(2.4442/0.10-0.06)}{(1.10)^2}$$

PV = 54.109 $

$\therefore$ Dividend yield $= \frac{D_1}{P_0}$

We take

Dividend yield $= \frac{1.92}{54.109}$

Dividend yield = 0.03548

CGY = total return − dividend yield

= 0.10 − 0.03548

= 0.06452

Capital gain yield = 6.452 %

Date:

b) Period of supernormal growth increased to another 5 years $g_1, g_2, g_3, g_4, g_5 = 20\%$ what happened to Div. yeild and CGY:-

→ If price goes lower:-
Div. yield will increase and CGY will decrease.

→ If price increases:-
Div yield decrease, CGY increase

c) what will happen to Div. yield and CGY if supernormal growth ends (constant growth)

% return = Div. yeild + CGY

% Dividend yeild = $r - g$ (constant growth)
= $10 - 6$ (CGY becomes growth)

Dividend yeild = 4 %

capital gain yield = growth (constant growth)

CGY = 6 %

d) Of what interest to investors is the relationship overtime b/w div. yield and CGY?

when growth slows down / becomes normal,
CGY ↓ (stock prices grows slowly
Div. yield ↑ (company starts paying more dividends)